Geographic Patterns in Diabetes and Related Risk Factors in the United States

Nanda Shiny Binu

HIT 723 – Public Health Informatics

WEEK 7 - Project Draft

1. Background Information on the Public Health Topic of Interest

Diabetes is an important public health concern because it can contribute to serious long term complications and requires ongoing prevention, early identification, and management. Type 2 diabetes is associated with several risk factors, including overweight or obesity and insufficient physical activity. Some risk factors, such as age and family history, cannot be changed, while healthy eating, increased physical activity, and weight management can help reduce risk for type 2 diabetes (Centers for Disease Control and Prevention [CDC], 2024a). Because risk and prevention opportunities differ across populations and communities, examining diabetes at the local level can help public health professionals identify where prevention resources may be needed.

This project focuses on geographic patterns in estimated diabetes prevalence and related risk factors across U.S. counties. The primary dataset is the CDC PLACES: Local Data for Better Health, County Data, 2025 release. PLACES provides local estimates for chronic disease and other health related measures across counties and other geographic areas in the United States. The 2025 release includes 40 measures, such as health outcomes, preventive practices, health risk behaviors, disability, health status, and health related social needs (CDC, 2025a). For this project, the main measure of interest is diabetes, considered alongside obesity, physical inactivity, smoking, high blood pressure, and high cholesterol where comparable measures are available.

The PLACES dataset is useful because local level health estimates are not always available from direct surveys for every county. PLACES applies small area estimation methods to information from the Behavioral Risk Factor Surveillance System (BRFSS), U.S. Census population data,

and American Community Survey estimates. Its multilevel regression and poststratification approach produce modeled estimates for local areas, including counties (CDC, 2025b). Research describing PLACES explains that this approach was developed to provide consistent local population health information that can support public health practice (Greenlund et al., 2022).

The modeled nature of the estimates is important when interpreting results. PLACES estimates describe population level patterns; they are not individual diagnoses and do not establish that one risk factor causes higher diabetes prevalence in a particular county. The CDC also cautions that the small area model cannot detect the effects of local interventions, so these estimates should not be used by themselves to evaluate a program or policy (CDC, 2025c). In this project, visual comparisons will therefore be described as geographic patterns or associations rather than proof of cause and effect.

Public health informatics connects these data to action by organizing, analyzing, and communicating health information in a form that supports decisions. Tableau maps and charts can make differences between counties easier to identify than a large spreadsheet. For example, a county map can display estimated diabetes prevalence, while bar charts can compare counties with higher and lower estimates for obesity or physical inactivity. These visualizations can help generate questions about local needs, guide further investigation, and communicate findings to audiences who may not work directly with data.

2. Possible Stakeholders

State and local health departments are key stakeholders because they plan and coordinate chronic disease prevention activities. County level estimates can help them identify communities that may benefit from additional outreach, diabetes screening initiatives, nutrition education, or

physical activity programs. PLACES is designed to help local health officials understand health problems and prioritize prevention activities (CDC, 2025a). The dashboard could serve as an initial screening tool for identifying geographic areas for closer review, rather than as the sole basis for allocating resources.

Clinicians, community health centers, and health systems may also benefit from local summaries. Understanding the estimated burden of diabetes and related risk factors in the communities they serve can support community health needs assessments and inform partnerships with prevention programs. These data should complement, not replace, clinical records, patient level assessment, or local surveillance information.

Public health planners and policymakers are another important audience. County comparisons can help them communicate the distribution of chronic disease burden, consider where prevention investments may be useful, and support discussions about access to health promoting resources. Because PLACES estimates are modeled and do not measure the effects of local interventions, decision makers should combine the dashboard with other evidence before judging program effectiveness or making high stakes decisions (CDC, 2025c).

Community based organizations, including groups that provide nutrition education, physical activity opportunities, diabetes education, and social services, may use the findings to understand community needs and coordinate outreach. Community members and the general public are also stakeholders because accessible maps can make local health patterns easier to understand and encourage informed conversations about prevention. The dashboard should use clear labels, plain language explanations, and notes about uncertainty and limitations so that users do not interpret county estimates as descriptions of every individual living in that county.

Researchers, students, and public health informatics professionals may use the dataset to explore geographic variation, develop visualizations, and identify questions for additional study. Their role includes checking measure definitions, documenting data sources, and interpreting patterns carefully. Across all stakeholder groups, responsible use requires transparent methods and recognition that county level averages can conceal differences among people and neighborhoods.

3. Possible Uses for Creating a Surveillance System Revolving Around This Topic

A diabetes focused surveillance dashboard could bring together PLACES estimates for diabetes and related risk factors in one interactive view. The initial system could allow users to select a county or state, compare diabetes prevalence across counties, and view related measures such as obesity and physical inactivity. A map would show the geographic distribution of estimated diabetes prevalence, while ranked bar charts could highlight counties with relatively higher or lower estimates. Filters for state and measure would help users explore the data without needing to manipulate the original dataset.

The first use would be to identify geographic patterns and potential priority areas. If some counties show higher estimated diabetes prevalence alongside higher estimates for obesity or physical inactivity, health departments could investigate whether these areas also face barriers to preventive care, healthy food, safe places to exercise, or diabetes education. The dashboard would not prove why a pattern exists, but it could help formulate questions and direct attention to other data sources. Local health departments could then combine the PLACES findings with community assessments, program participation data, healthcare access measures, and input from residents.

A second use would be planning and targeting prevention activities. Public health teams could use the map to support discussions about where to expand diabetes prevention education, partner with community organizations, or conduct outreach about screening and lifestyle programs. Clinicians and community health organizations could use county level information as background for community health needs assessments. The data could also support public facing communication by presenting local patterns in a clear, consistent format.

A third use would be ongoing situational awareness and reporting. When new PLACES releases become available, the dashboard could be updated using the corresponding release and its documented definitions. Users could review whether the geographic distribution of estimates appears different between releases, while recognizing that changes may reflect updated source data or methods as well as changes in population health. Because PLACES estimates are not designed to evaluate the effects of local interventions, the dashboard should not be used alone to claim that a program caused a change (CDC, 2025c). Any comparison over time should document the release years and check whether the measures and methods are comparable.

To support trustworthy surveillance, the system should include the data source, release year, measure definitions, geographic level, and a short explanation of modeled estimates. It should distinguish estimated prevalence from counts of diagnosed individuals and avoid presenting county rankings as definitive judgments about community performance. The system should also be supplemented with local knowledge and additional data before decisions are made. In this way, the dashboard would function as a practical public health informatics tool: it would transform population level data into understandable information, support prioritization and communication, and help stakeholders identify where further investigation or prevention planning may be warranted.

4. References

Centers for Disease Control and Prevention. (2024a, May 15). *Diabetes risk factors*. https://www.cdc.gov/diabetes/risk-factors/

Centers for Disease Control and Prevention. (2025a, December 7). *About PLACES: Local Data for Better Health*. https://www.cdc.gov/places/about/

Centers for Disease Control and Prevention. (2025b, December 8). *Methodology*. https://www.cdc.gov/places/methodology/

Centers for Disease Control and Prevention. (2025c, September 22). *PLACES: Local Data for Better Health, County Data, 2025 release*. https://data.cdc.gov/500-Cities-Places/PLACES-Local-Data-for-Better-Health-County-Data-20/swc5-untb

Greenlund, K. J., Lu, H., Wang, Y., Matthews, K. A., LeClercq, J. M., Lee, B., & Carlson, S. A. (2022). PLACES: Local data for better health. *Preventing Chronic Disease, 19*, Article E31. https://doi.org/10.5888/pcd19.210459